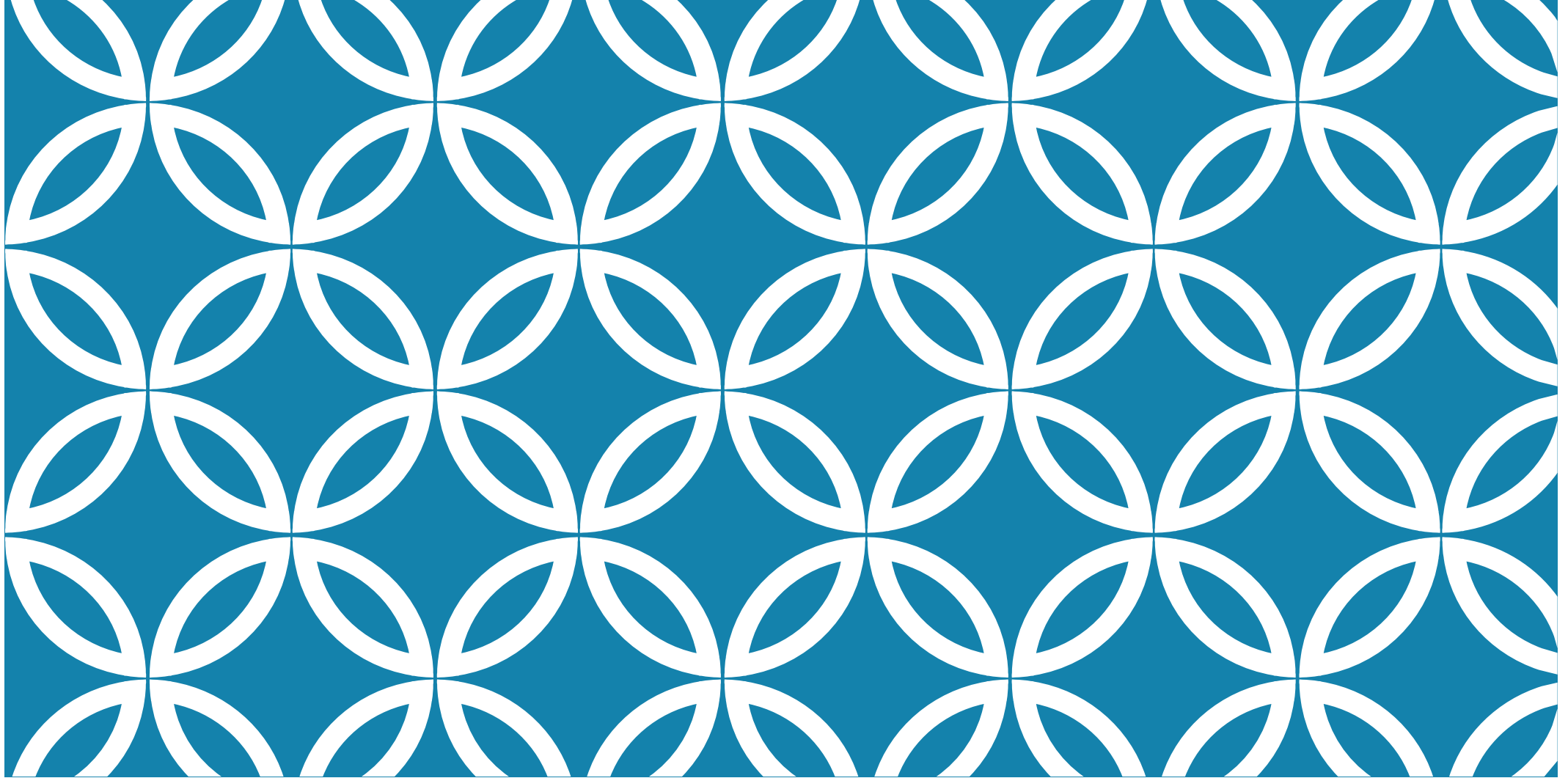




Business Intelligence: Introduction to Business Intelligence

Lecture 1

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LEARNING OBJECTIVES

Understand today's turbulent business environment and describe how organizations survive and even excel in such an environment (solving problems and exploiting opportunities)

Understand the need for computerized support of managerial decision making

Describe the business intelligence (BI) methodology and concepts and relate them to decision support systems (DSS)

Understand the issues in implementing BI

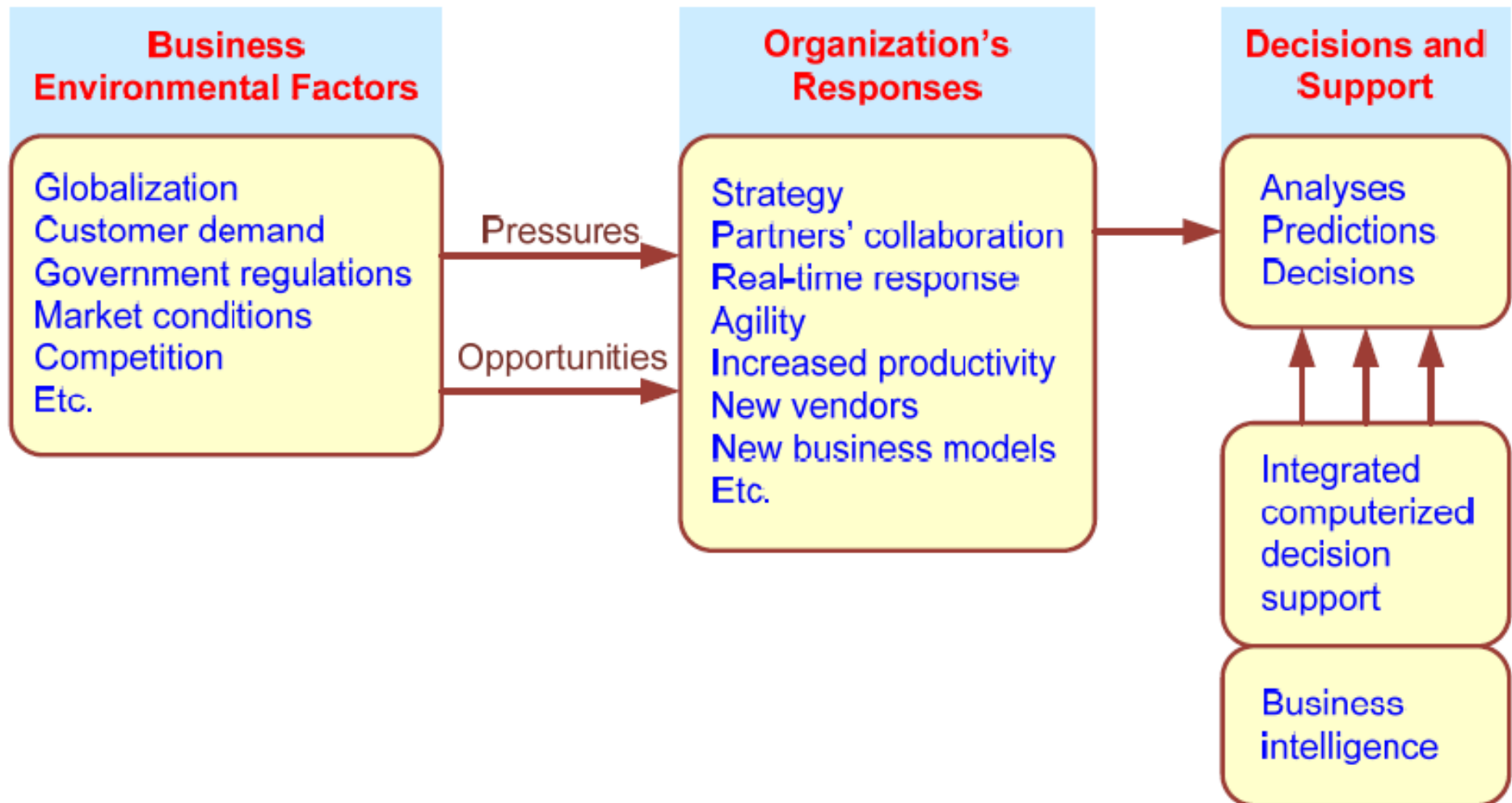
CHANGING BUSINESS ENVIRONMENT & COMPUTERIZED DECISION SUPPORT

Companies are moving aggressively to computerized support of their operations => Business Intelligence

Business Pressures—Responses—Support Model

- **Business pressures** result of today's competitive business climate
- **Responses** to counter the pressures
- **Support** to better facilitate the process

BUSINESS PRESSURES—RESPONSES—SUPPORT MODEL



THE BUSINESS ENVIRONMENT

The environment in which organizations operate today is becoming more and more complex, creating:

- opportunities, and
- problems.
- Example: globalization.

Business environment factors:

- markets, consumer demands, technology, and societal.

BUSINESS ENVIRONMENT FACTORS

FACTOR

DESCRIPTION

Markets

Strong competition
Expanding global markets
Blooming electronic markets on the Internet
Innovative marketing methods
Opportunities for outsourcing with IT support
Need for real-time, on-demand transactions

Consumer demand

Desire for customization
Desire for quality, diversity of products, and speed of delivery
Customers getting powerful and less loyal

Technology

More innovations, new products, and new services
Increasing obsolescence rate
Increasing information overload
Social networking, Web 2.0 and beyond

Societal

Growing government regulations and deregulation
Workforce more diversified, older, and composed of more women
Prime concerns of homeland security and terrorist attacks
Necessity of Sarbanes-Oxley Act and other reporting-related legislation
Increasing social responsibility of companies
Greater emphasis on sustainability

ORGANIZATIONAL RESPONSES

Be Reactive, Anticipative, Adaptive, and Proactive

Managers may take actions, such as:

- Employing strategic planning.
- Using new and innovative business models.
- Restructuring business processes.
- Participating in business alliances.
- Improving corporate information systems.
- Improving partnership relationships.
- Encouraging innovation and creativity. ...cont...>

ORGANIZATIONAL RESPONSES, CONTINUED

- Improving customer service and relationships.
- Moving to electronic commerce (e-commerce).
- Moving to make-to-order production and on-demand manufacturing and services.
- Using new IT to improve communication, data access (discovery of information), and collaboration.
- Responding quickly to competitors' actions (e.g., in pricing, promotions, new products and services).
- Automating many tasks of white-collar employees.
- Automating certain decision processes.
- Improving decision-making by employing analytics.

CLOSING THE STRATEGY GAP

One of the major objectives of computerized decision support is to facilitate closing the gap between the current performance of an organization and its desired performance, as expressed in its mission, objectives, and goals, and the strategy to achieve them.

BUSINESS INTELLIGENCE (BI)

BI is an evolution of decision support concepts over time.

- Meaning of EIS/DSS...
 - **Then:** Executive Information System
 - **Now:** Everybody's Information System (BI)

BI systems are enhanced with additional visualizations, alerts, and performance measurement capabilities.

The term BI emerged from industry apps.

DEFINITION OF BI

BI is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies.

BI a content-free expression, so it means different things to different people.

BI's major objective is to enable easy access to data (and models) to provide business managers with the ability to conduct analysis.

BI helps *transform* data, to information (and knowledge), to decisions and finally to action.

A BRIEF HISTORY OF BI

The Gartner Group coined the term BI in the mid-1990s

However, the concept is much older

- 1970s — MIS reporting — static/periodic reports
- 1980s — Executive Information Systems (EIS)
- 1990s — OLAP, dynamic, multidimensional, ad-hoc reporting -> coining of the term “BI”
- 2005+ — Inclusion of AI and Data/Text Mining capabilities; Web-based Portals/Dashboards
- 2010s — Yet to be seen

THE EVOLUTION OF BI CAPABILITIES

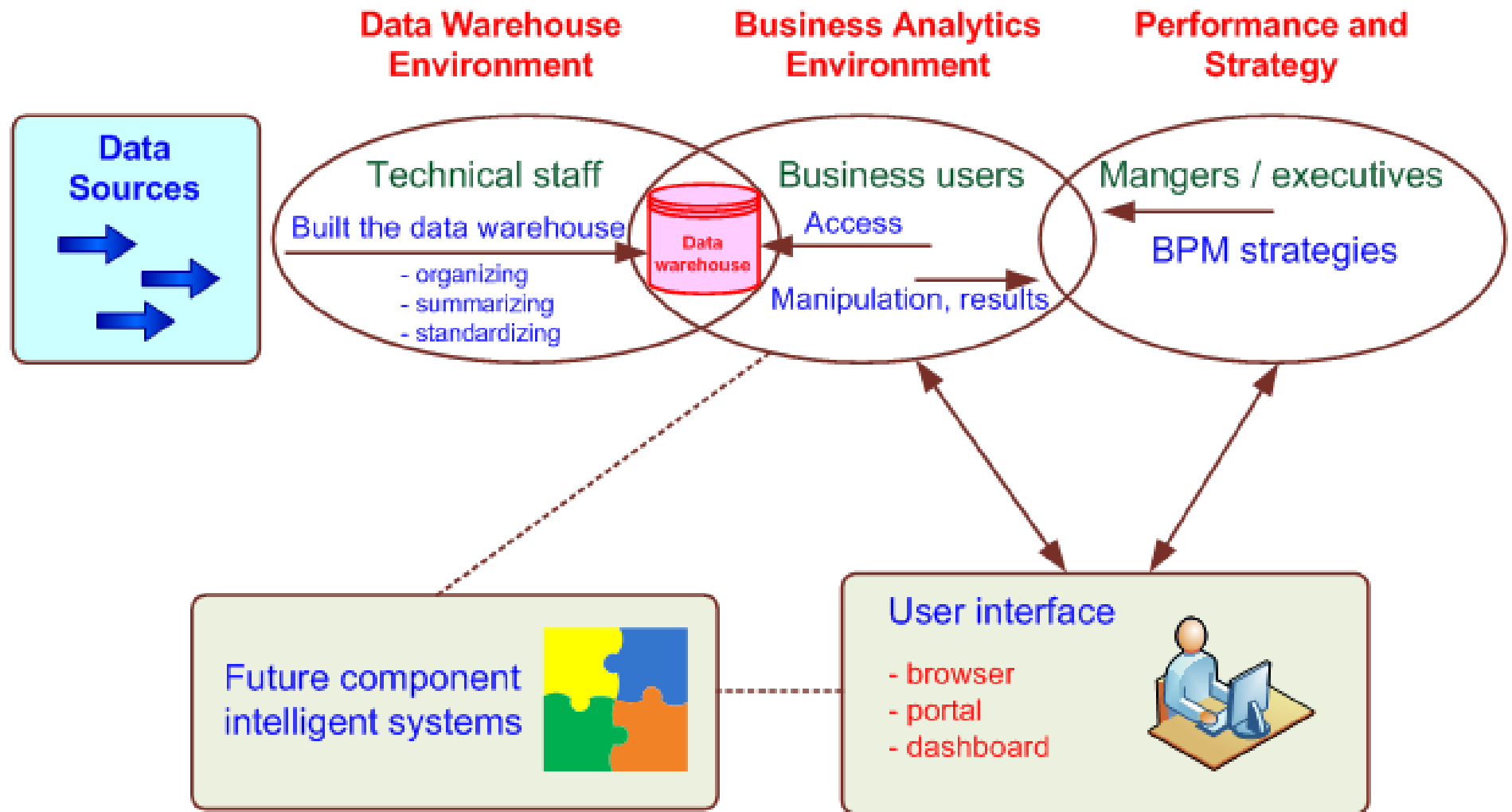


THE ARCHITECTURE OF BI

A BI system has four major components:

- a data warehouse, with its source data
- business analytics, a collection of tools for manipulating, mining, and analyzing the data in the data warehouse;
- business performance management (BPM) for monitoring and analyzing performance
- a user interface (e.g., dashboard)

A HIGH-LEVEL ARCHITECTURE OF BI



COMPONENTS IN A BI ARCHITECTURE

The **data warehouse** is the cornerstone of any medium-to-large BI system.

- Originally, the data warehouse included only historical data that was organized and summarized, so end users could easily view or manipulate it.
- Today, some data warehouses include access to current data as well, so they can provide real-time decision support (for details see Chapter 2).

Business analytics are the tools that help users transform data into knowledge (e.g., queries, data/text mining tools, etc.).

BI EXAMPLES

Epagogix is an analytics based BI system that specializes in predicting success of movies based on a detailed analysis of movie scripts.

National Australia Bank uses data mining to aid its marketing initiatives.

Hoyt Highland Partners, a marketing intelligence firm, assists health care providers with growing their businesses.

COMPONENTS IN A BI ARCHITECTURE

Business Performance Management (BPM), which is also referred to as corporate performance management (CPM), is an emerging portfolio of applications within the BI framework that provides enterprises tools they need to better manage their operations (for details see Chapter 3).

User Interface (i.e., dashboards) provides a comprehensive graphical/pictorial view of corporate performance measures, trends, and exceptions.

THE BENEFITS OF BI

The ability to provide accurate information when needed, including a real-time view of the corporate performance and its parts

A survey by Thompson (2004)

- Faster, more accurate reporting (81%)
- Improved decision making (78%)
- Improved customer service (56%)
- Increased revenue (49%)

See Table 1.2 for a list of **BI analytic applications**, the **business questions** they answer and the **business value** they bring.

TRANSACTION PROCESSING VERSUS ANALYTIC PROCESSING

Transaction processing systems are constantly involved in handling updates (add/edit/delete) to what we might call operational databases.

- ATM withdrawal transaction, sales order entry via an ecommerce site – updates DBs
- Online analytic processing (OLTP) handles routine on-going business
- ERP, SCM, CRM systems generate and store data in OLTP systems
- The main goal is to have high efficiency

TRANSACTION PROCESSING VERSUS ANALYTIC PROCESSING

Online analytic processing (OLAP) systems are involved in extracting information from data stored by OLTP systems

- Routine sales reports by product, by region, by sales person, etc.
- Often built on top of a data warehouse where the data is not transactional
- Main goal is effectiveness (and then, efficiency) – provide correct information in a timely manner
- More on OLAP will be covered in Chapter 2

SUCCESSFUL BI IMPLEMENTATION

Implementing and deploying a BI initiative is a lengthy, expensive and risky endeavor!

Success of a BI system is measured by its widespread usage for better decision making.

The typical BI user community includes

- All levels of the management hierarchy (not just the top executives, as was for EIS)
 - Provide what is needed to whom he/she needs it

A successful BI system must be of benefit to the enterprise as a whole.

BI AND BUSINESS STRATEGY

To be successful, BI must be aligned with the company's business strategy.

- BI cannot/should not be a technical exercise for the information systems department.

BI changes the way a company conducts business by

- improving business processes, and
- transforming decision making to a more data/fact/information driven activity.

BI should help execute the business strategy and not be an impediment for it!

REAL-TIME, ON-DEMAND BI

The demand for “real-time” BI is growing!

Is “real-time” BI attainable?

Technology is getting there...

- Automated, faster data collection (RFID, sensors,...)
- Database and other software technologies (agent, SOA, ...) are advancing
- Telecommunication infrastructure is improving
- Computational power is increasing while the cost for these technologies is decreasing

Trent -> Business Activity Management

ISSUES FOR SUCCESSFUL BI

Developing vs. Acquiring BI systems

- Developing everything from scratch
- Buying/leasing a complete system
- Using a shell BI system and customizing it
- Use of outside consultants?

Justifying via cost-benefit analysis

- It is easier to quantify costs
- Harder to quantify benefits
 - Most of them are intangibles

ISSUES FOR SUCCESSFUL BI

Security and Privacy

- Still an important research topic in BI
- How much security/privacy?

Integration of Systems and Applications

- BI must integrate into the existing IS
 - Often sits on top of ERP, SCM, CRM systems
- Integration to outside (partners of the extended enterprise) via internet –
 - customers, vendors, government agencies, etc.

MAJOR BI TOOLS AND TECHNIQUES

Tool categories

- Data management
- Reporting, status tracking
- Visualization
- Strategy and performance management
- Business analytics
- Social networking & Web 2.0
- New/advanced tools/techniques to handle massive data sets for knowledge discovery